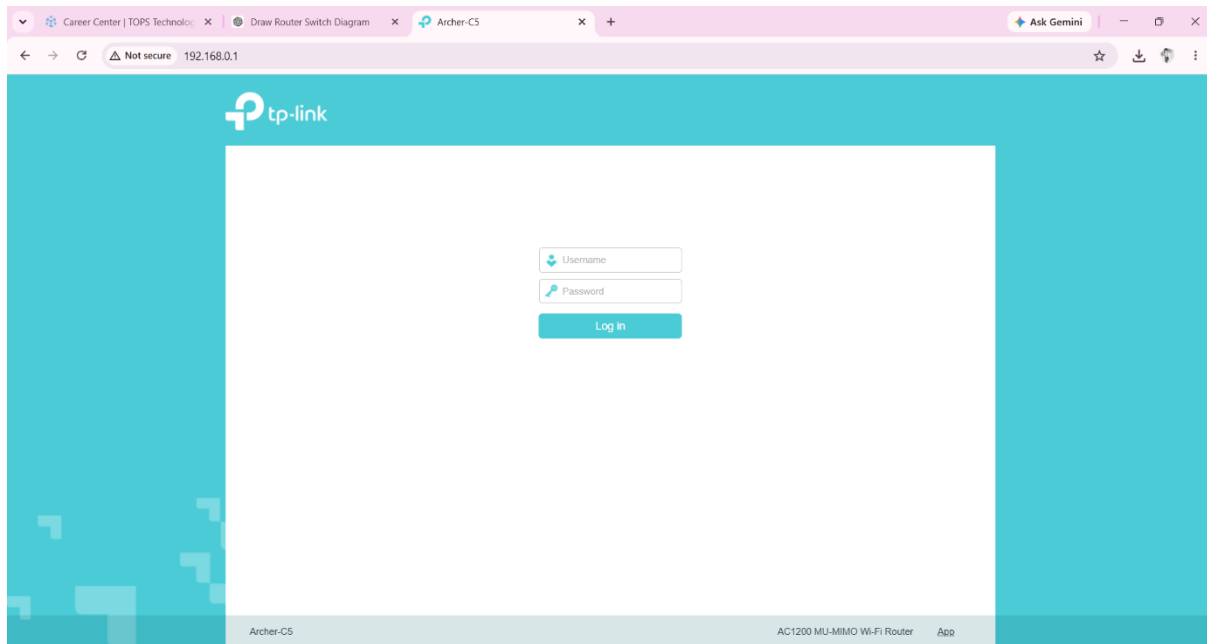


Day 3 : Configuring Basic Network HardwareTheory

Task 1 : Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

- First, I checked the Default Gateway using the ipconfig command in Command Prompt. I entered the Default Gateway address in a web browser to open the router login page.
- My Default Gateway is : 192.168.0.1



Task 2 : Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

- I used the ipconfig command to check the current network details of my laptop.

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix  . : 
IPv6 Address. . . . . : 2406:b400:f11:e9f7:1b00:f970:c3a8:329b
Temporary IPv6 Address. . . . . : 2406:b400:f11:e9f7:b922:272:3dcc:17c2
Link-local IPv6 Address . . . . . : fe80::7c57:b381:11:51d%8
IPv4 Address. . . . . : 192.168.0.11
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::bafb:b3ff:fea2:4a1b%8
                          192.168.0.1
```

Network Detail	Value
IPv4 Address	192.168.0.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1

Task 3 : Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

- I changed the network settings from automatic DHCP to a static IP address. After applying the settings, I checked the Internet connection to confirm that the configuration was working.

Edit IP settings

IPv4

☒ On

IP address

192.168.0.50

Subnet mask

255.255.255.0

Gateway

192.168.0.1

Preferred DNS

8.8.8.8

DNS over HTTPS

Off

Alternate DNS

8.8.4.4

Save

Cancel

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :  
IPv6 Address. . . . . : 2406:b400:f11:e9f7:1b00:f970:c3a8:329b  
Temporary IPv6 Address. . . . . : 2406:b400:f11:e9f7:b922:272:3dcc:17c2  
Link-local IPv6 Address . . . . . : fe80::7c57:b381:11:51d%8  
IPv4 Address. . . . . : 192.168.0.50  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::bafb:b3ff:fea2:4a1b%8  
                            192.168.0.1
```

The Internet connection was tested successfully. After testing, the settings were changed back to automatic DHCP.

Task 4 : In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

DHCP Range and IP Address Conflict

Objective

To locate the DHCP settings in the router's administration panel, identify the current DHCP IP address range, and understand what happens if two devices are assigned the same IP address.

Procedure

1. Logged in to the router's **Administration Panel** using a web browser.
2. Opened the **DHCP Settings** section.
3. Confirmed that **DHCP is enabled** and **DHCP Server** mode is selected.
4. Located the **IP Address Pool**, which defines the range of IP addresses that the DHCP server can automatically assign to devices.
5. Recorded the current DHCP range and other relevant settings.

Current DHCP Configuration

Setting	Current Value
DHCP	Enabled
DHCP Mode	DHCP Server
IP Address Pool	192.168.0.2 – 192.168.0.249
Address Lease Time	120 minutes
Default Gateway	0.0.0.0 (Optional)
Default Domain	Not configured

DHCP Range

The router's current DHCP IP address pool is:

Start IP: 192.168.0.2

End IP: 192.168.0.249

Therefore, the router can automatically assign IP addresses from **192.168.0.2 to 192.168.0.249** to devices requesting an address through DHCP.

What Happens if Two Devices Have the Same IP Address?

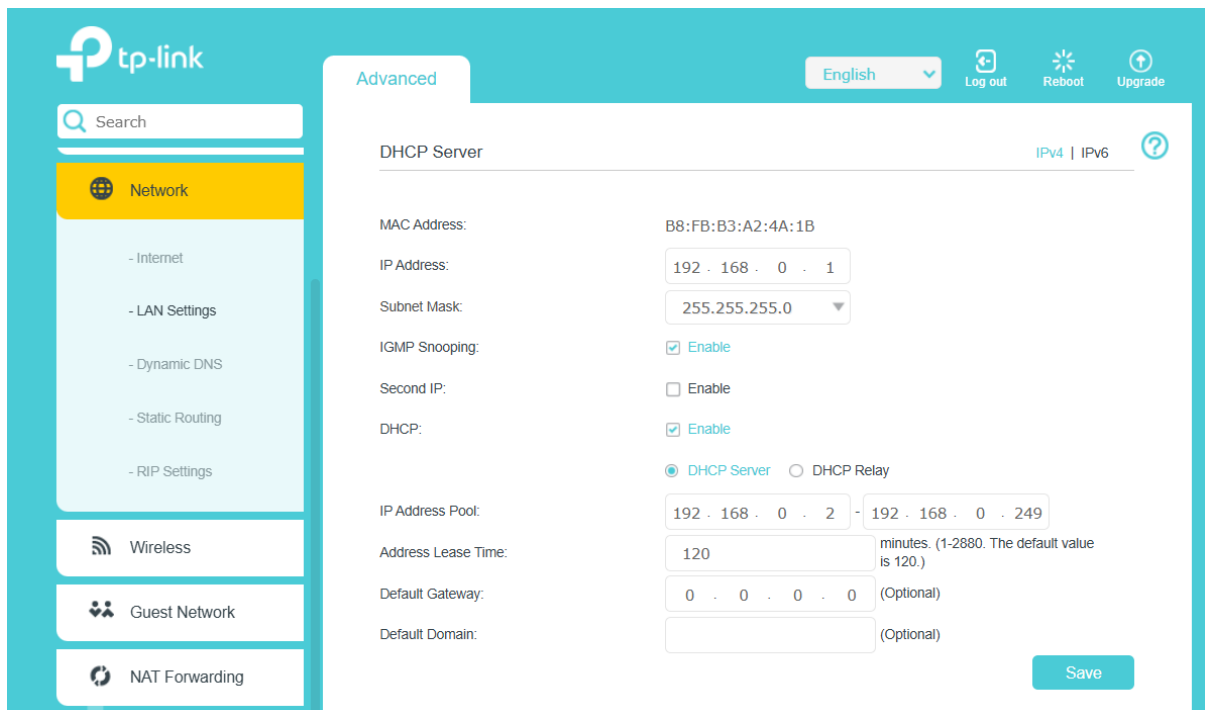
If two devices are assigned the same IP address, an **IP address conflict** occurs because the network cannot uniquely identify the two devices. This can cause **intermittent connectivity, loss of Internet access, or problems communicating with other devices** on the network. Normally, the DHCP server keeps track of leased addresses to avoid assigning the same IP address to two devices at the same time.

Result

The DHCP settings were successfully located in the router's administration panel. The DHCP server is enabled, and the configured IP address pool is **192.168.0.2–192.168.0.249** with a **120-minute lease time**.

Evidence

Router DHCP Settings Screenshot:



The screenshot shows the TP-Link router's administration interface. On the left is a sidebar with a search bar and a menu containing 'Network' (highlighted), 'Wireless', 'Guest Network', and 'NAT Forwarding'. The 'Network' section is expanded, showing sub-items: '- Internet', '- LAN Settings', '- Dynamic DNS', '- Static Routing', and '- RIP Settings'. The main content area is titled 'Advanced' and 'DHCP Server'. It features a language dropdown set to 'English' and buttons for 'Log out', 'Reboot', and 'Upgrade'. The DHCP settings are for IPv4. The MAC Address is B8:FB:B3:A2:4A:1B. The IP Address is 192.168.0.1. The Subnet Mask is 255.255.255.0. IGMP Snooping is enabled. The Second IP is disabled. DHCP is enabled. The DHCP Server radio button is selected over the DHCP Relay option. The IP Address Pool is 192.168.0.2 to 192.168.0.249. The Address Lease Time is 120 minutes. The Default Gateway is 0.0.0.0. The Default Domain is empty. A 'Save' button is at the bottom right.

Setting	Value
MAC Address:	B8:FB:B3:A2:4A:1B
IP Address:	192.168.0.1
Subnet Mask:	255.255.255.0
IGMP Snooping:	☒ Enable
Second IP:	☐ Enable
DHCP:	☒ Enable
	☒ DHCP Server ☐ DHCP Relay
IP Address Pool:	192.168.0.2 - 192.168.0.249
Address Lease Time:	120 minutes. (1-2880. The default value is 120.)
Default Gateway:	0.0.0.0 (Optional)
Default Domain:	(Optional)

Conclusion

The router's DHCP server automatically provides IP addresses to devices connected to the network within the configured DHCP pool. Proper DHCP management helps prevent IP address conflicts and ensures reliable communication between devices on the network.